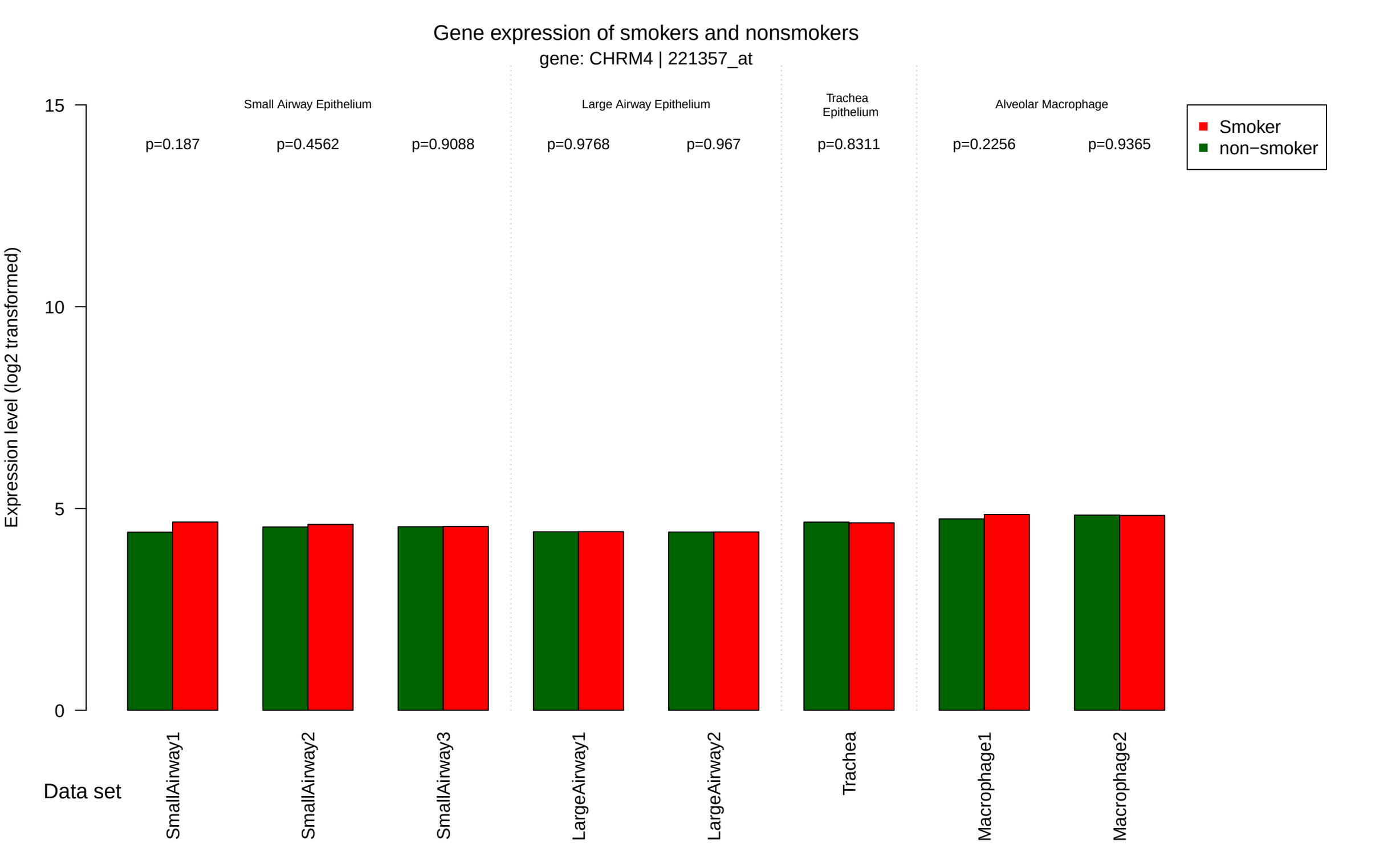
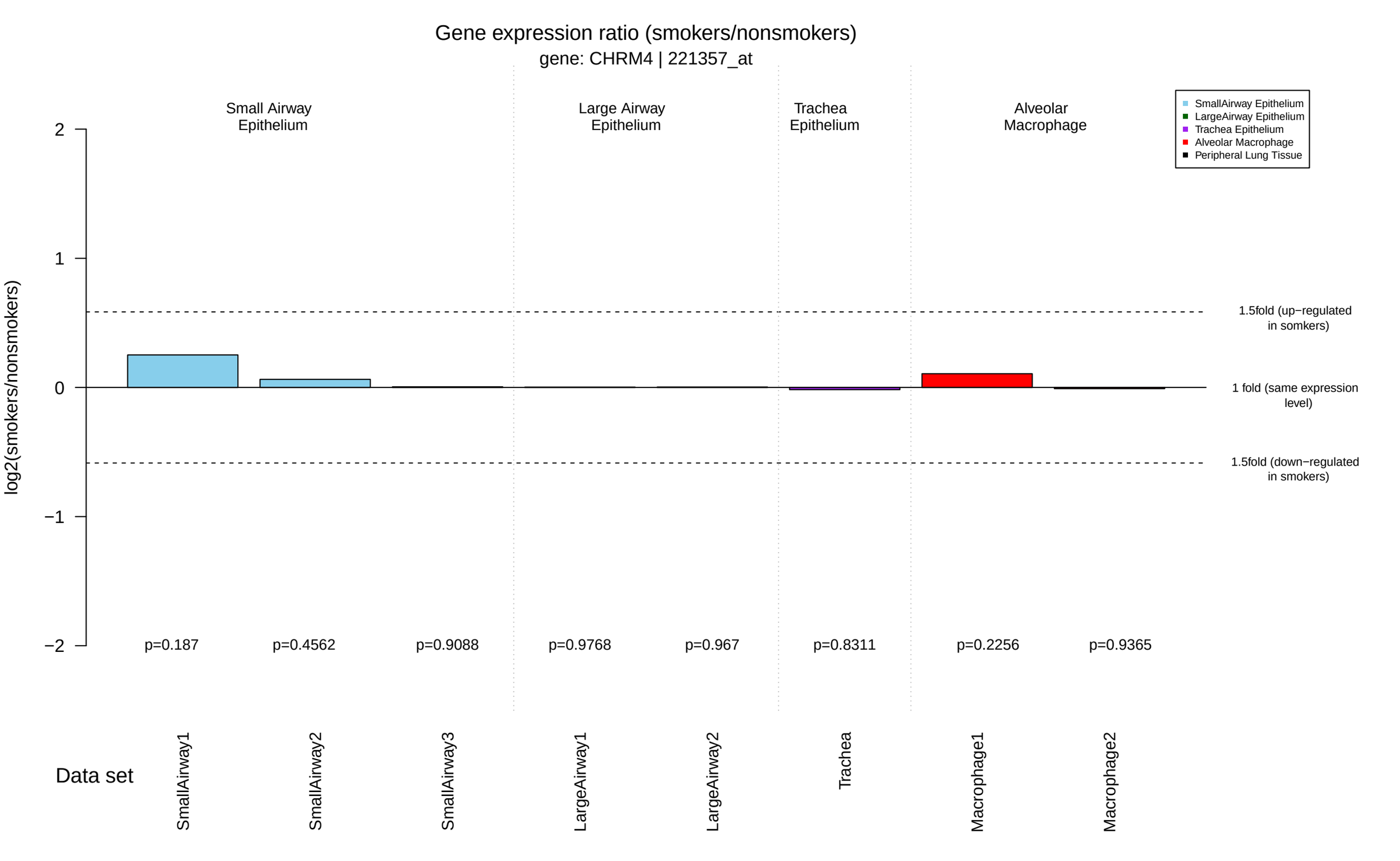
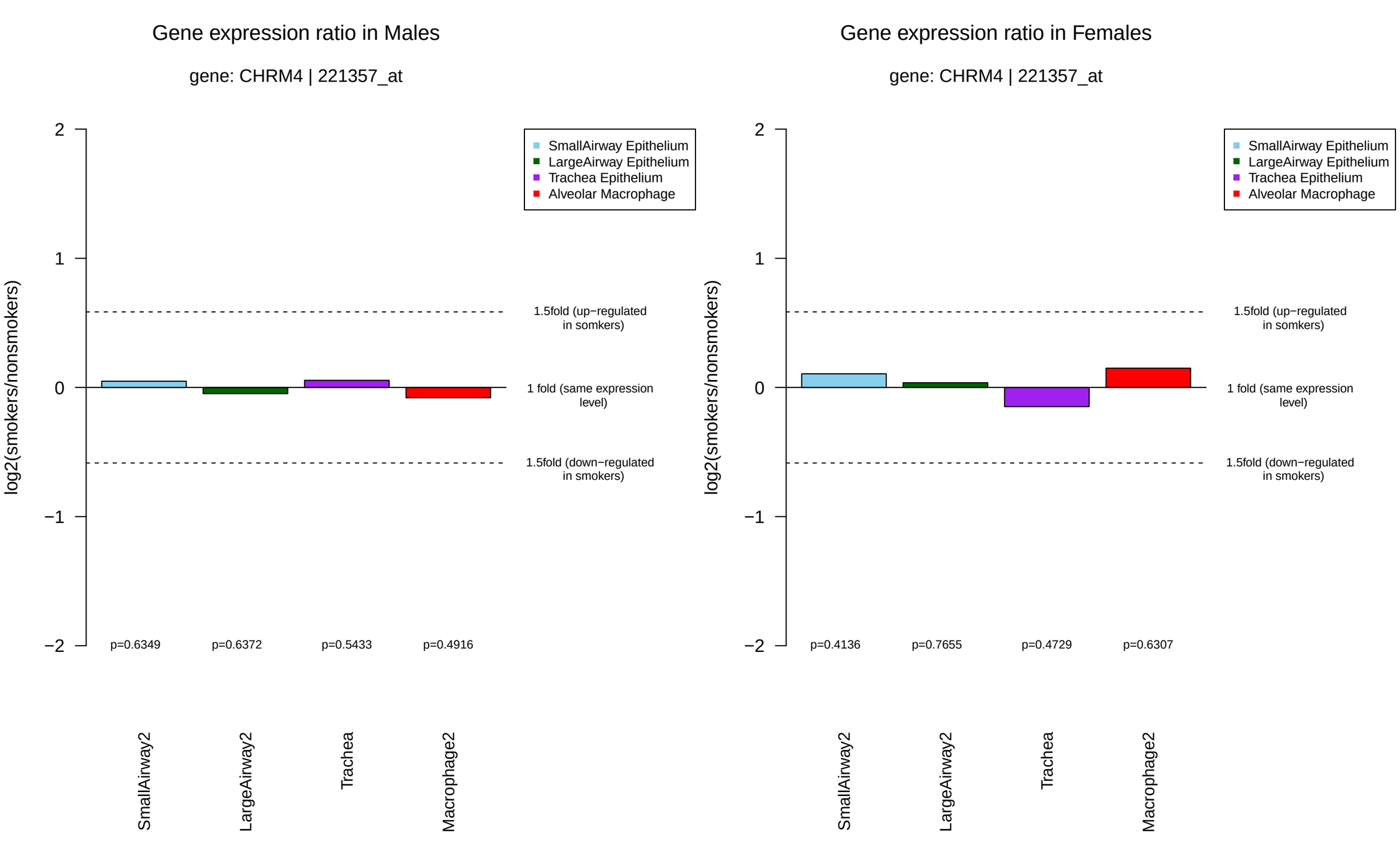
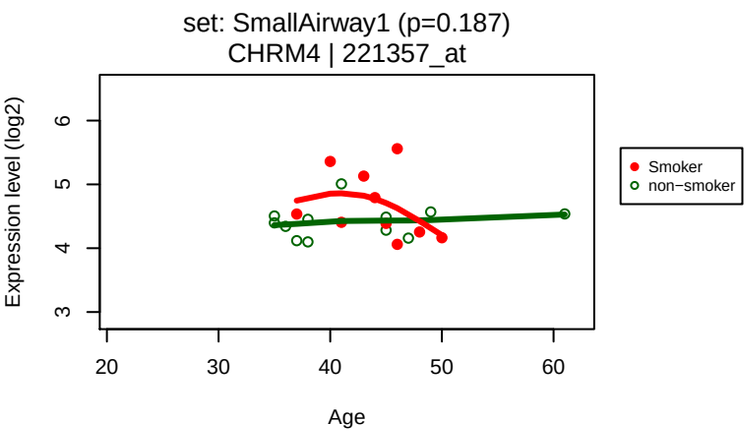
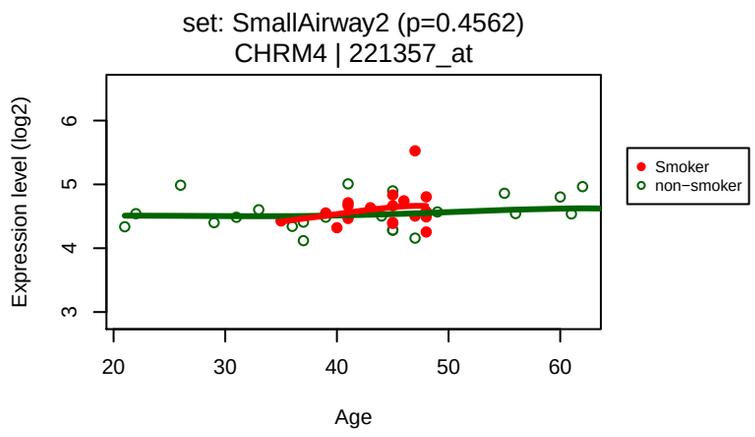
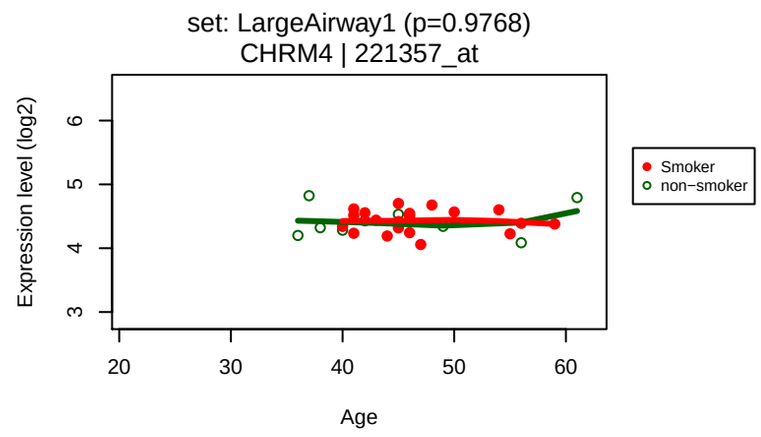

Gene expression of smokers and nonsmokers
gene: CHRM4 | 221357_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.187
p=0.4562
p=0.9088
p=0.9768
p=0.967
p=0.8311
p=0.2256
p=0.9365
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

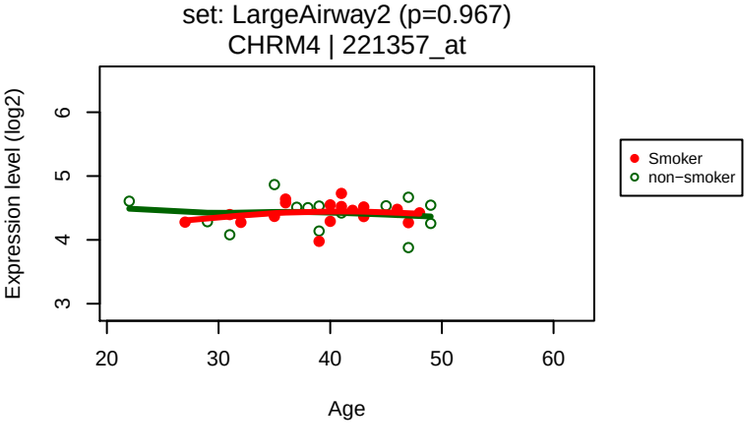

Gene expression ratio (smokers/nonsmokers)
gene: CHRM4 | 221357_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.187
p=0.4562
p=0.9088
p=0.9768
p=0.967
p=0.8311
p=0.2256
p=0.9365
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

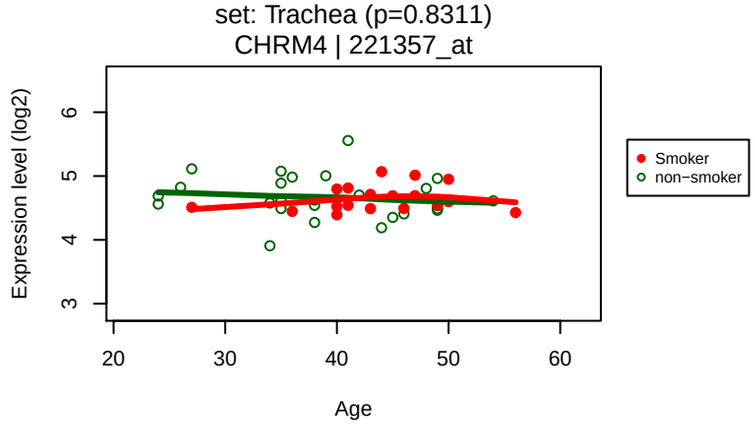

Gene expression ratio in Males
gene: CHRM4 | 221357_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6349
p=0.6372
p=0.5433
p=0.4916
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRM4 | 221357_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4136
p=0.7655
p=0.4729
p=0.6307
SmallAirway2
LargeAirway2
Trachea
Macrophage2

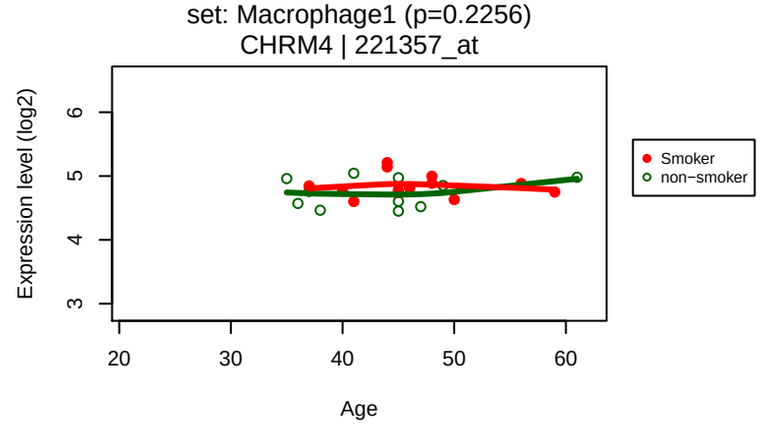

set: SmallAirway1 (p=0.187)
CHRM4 | 221357_at
Expression level (log2)
Age
Smoker
non−smoker

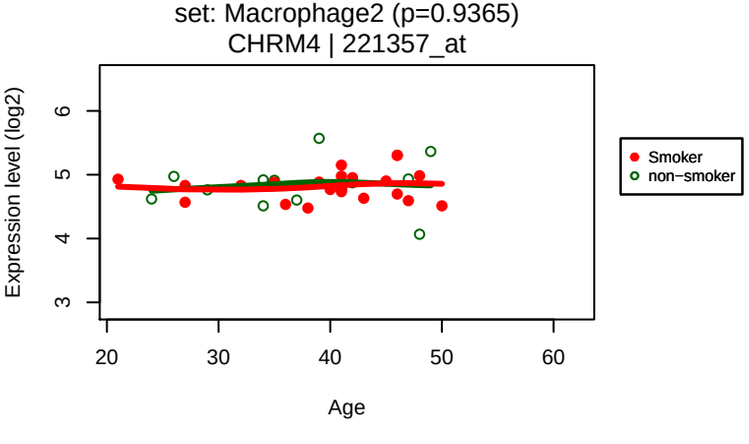

set: SmallAirway2 (p=0.4562)
CHRM4 | 221357_at
Expression level (log2)
Age
Smoker
non−smoker

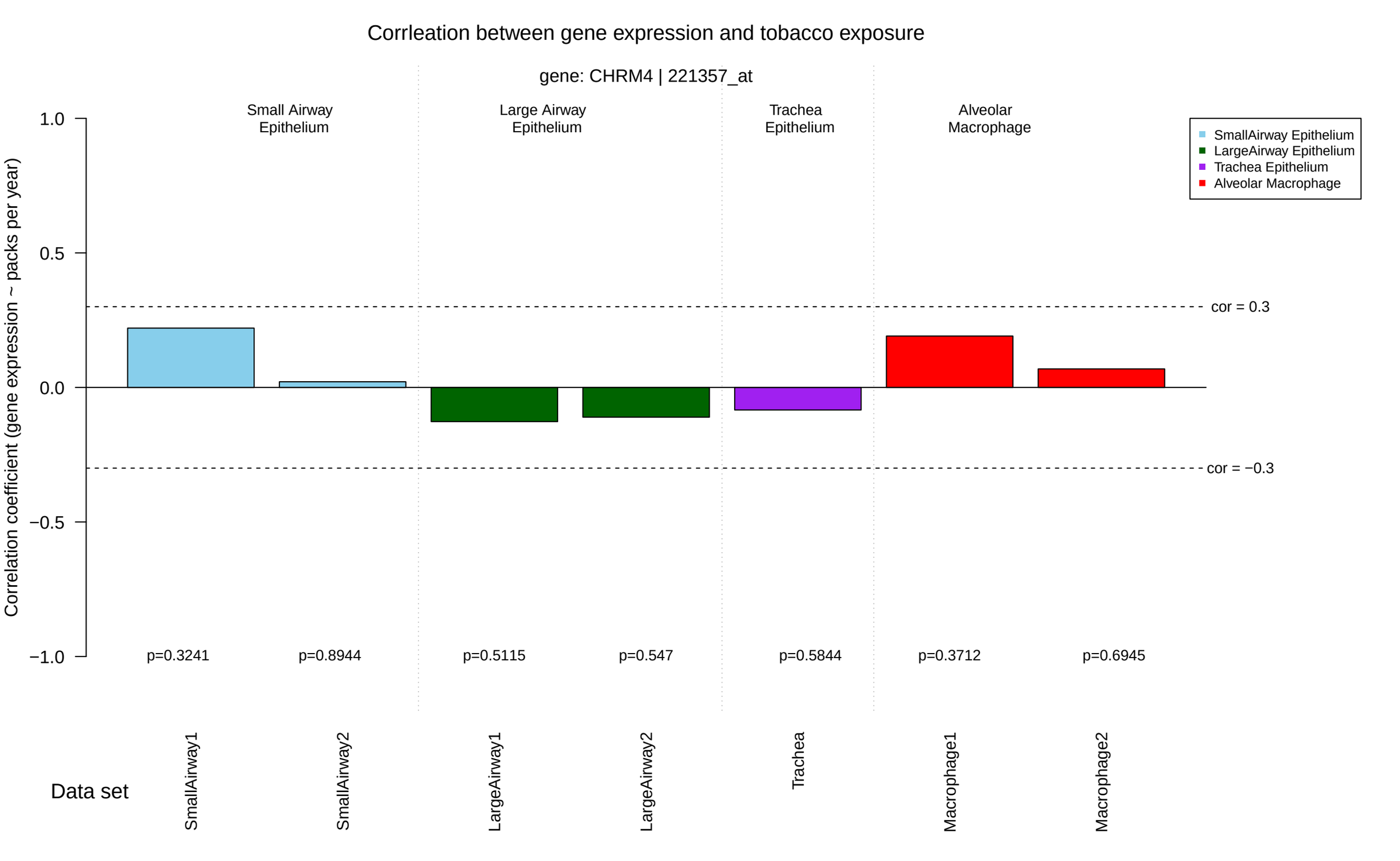

set: LargeAirway1 (p=0.9768)
CHRM4 | 221357_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.967)
CHRM4 | 221357_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8311)
CHRM4 | 221357_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2256)
CHRM4 | 221357_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9365)
CHRM4 | 221357_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRM4 | 221357_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3241
p=0.8944
p=0.5115
p=0.547
p=0.5844
p=0.3712
p=0.6945
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2